



කෘෂි කර්ම දෙපාර්තමේන්තුව
விவசாயத் திணைக்களம்
Department of Agriculture



(<https://doa.gov.lk/>)



දළසාත බෝග පර්යේෂණ සහ සංවර්ධන
பூங்கனியியல் பயிர் ஆராய்ச்சி அபிவிருத்தி
Horticultural Crop Research and Development

100 ruwa rd, Peradeniya, Sri Lanka

[a.gov.lk\(mailto:director.hordi@doa.gov.lk\)](mailto:director.hordi@doa.gov.lk)

8011-12-13

[://doa.gov.lk/sinhala-hordi-crop-niwithi\)](https://doa.gov.lk/sinhala-hordi-crop-niwithi)

தமிழ் (<https://doa.gov.lk/tamil-hordi-c>

[<< Crop Page \(/hordi-crops/\)](/hordi-crops/)



Niwithi

Spinacia oleracea

A green leafy vegetable rich in Iron. It contains oxalic acid which prevents the body from absorbing Calcium. Therefore, the calcium in the blood is not absorbed into the bones and may accumulate in the kidneys and bladder and deposited there.

Available Types

There are two types of spinach, erect type and vine type, depending on the growth habit. Large leafy vines are used for cultivation. There is a red spinach type that is edible. It's not so popular.

- Large vine type with green leaves (giant spinach)
 - Red spinach
-

Climatic requirements/ Areas suitable for cultivation

Can be grown in any part of Sri Lanka except in low temperature regions

Soil

Spinach needs well-drained places for cultivation.

Seed requirement

10 – 15 grams of seeds is required per 10m² bed. (10-15g / 10m² bed)

Field preparation and planting

Weeds should be controlled and the ground should be ploughed well and the soil should be finely prepared. It is important to improve drainage by placing raised beds in areas where water logged. It is advisable to plant two seeds in one place. It takes about 5-6 days for the seeds to germinate.

Planting material

Seeds are used for planting. Although cuttings can be used for planting, the early flowering will retard the growth of the plant. Therefore cuttings are not recommended as planting materials.

Providing support

If the cultivation is going on for a long time, 4 weeks after planting, prepare a trellis about 1 – 1 1/2 feet above the ground level.

Spacing

Erect type- The spacing between rows and the within rows is 15cm x 15cm.

Vines – 45cm between rows 45 cm within rows

Seeds and seedlings are used for planting.

Fertilizer**Organic fertilizer**

Poultry manure is widely used. Although cattle manure is more suitable, it increases the chances of weed growth. It can be used in home gardening. 25 kgs of poultry manure can be applied for 10m² bed. (25kgs/ 10m²) It is advisable to mix organic manure with the soil, few days before planting the seeds.

Chemical fertilizer

The amount of fertilizer to be applied per 100 square meters of land

Fertilizer application	Urea kg.	TSP kg.	MOP kg.
Basic fertilizer	1.0	1.5	1.0
2–3 weeks after planting	1.0	–	–

Note 1: It is advisable to apply phosphorus and potassium as basic fertilizers after a soil test.

Water supply/Irrigation

Irrigate the crop well every 3-4 days. In dry weather it is advisable to irrigate the crop more often without allowing them to wither.

Weed Management

Weeds should be removed and the field should be kept clean as long as the crop remains.

Disease Management

▲ [1. Anthracnose](#)

Causal Organism: *colletotrichum spinaciae*

Symptoms

- In initial stage – Small , circular water – soaked lesions on both young and old leaves

- In late stage – Lesion turns brown to brownish color and become thin and papery , Tiny black fruiting bodies (Acervuli) from profusely in diseased tissue and are a characteristic feature

Management

- Removal of diseased plant debris from the field
- Establishment of suitable drainage systems
- Sprinklers can be caused to increase spreading of disease, due to this reason it is suitable to use another favorable water supply system

▲ [2. Bacterial leaf spot](#)

Causal Organism: *Pseudomonas syringae pv.spinaciae*

Symptoms

Initial symptom is consist of water –soaked irregularly shaped spots that measure 0.12 -0.25 inch in diameter. As the disease develops they enlarge as much as 0.5 -0.75 inch in diameter, are angular and turn dark brown in color. Occasionally spots have block sections or edges. Sometimes these spots merge together and resulting the death of large areas of leaves can be seen in both newly expanded and mature foliage

Management

- Removal of affected leaves
- Don't use sprinkler irrigation system

▲ [3. Cladosporium leaf spot](#)

Causal Organism: *cladosporium variable*

Symptoms

Rarely exceed spots 0.25 inch in diameter. They are round and tan in colour. In final stage can be seen holes

Management

Treatment is rarely needed. Copper fungicides can be used when needed.

▲ [4. Damping – off and Root rot](#)

Causal Organism: *Fusarium oxysporum*, *phythium spp*, *rhizoctonia solani*

Symptoms

In nursery period wilting of seedlings and death of seedlings can be caused due to root rot.

Management

- Can be control by seed treatments, sterilization of nurseries and water management
- Application of fungicide if necessary

Harvesting

Harvesting can be done in two ways.

- After about 40-50 days, the seedlings are removed and taken as a harvest.
- After about 40-45 days, cut the upper part of the stem to a height of about 15 cm from the ground level and allow the bushes to grow again and harvest about 6-8 times.

Harvest

1000 – 1500 kg per square meter. (1000-1500 kg / m²)